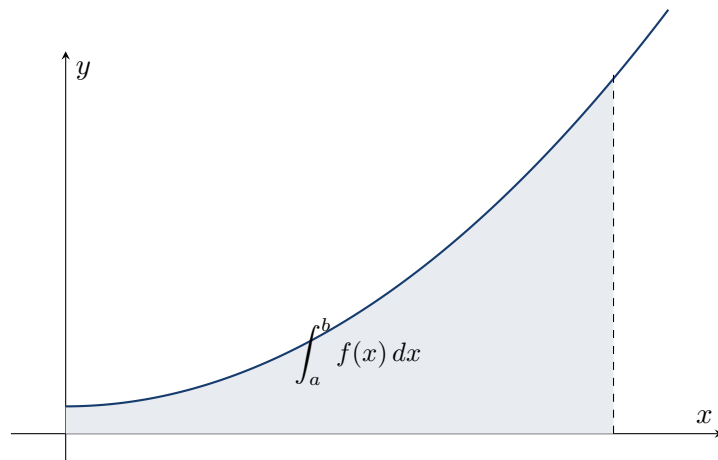


Integral Calculus

A Complete Exam-Prep Study Guide

For CSE 1-1 Students



Topics Covered:

- Riemann Sums & Definite Integrals
- Fundamental Theorem of Calculus
- Net Change Theorem
- Properties of Integrals
- Antiderivatives & Indefinite Integrals
- Substitution Rule

Includes worked examples, practice problems, full answer key, and important exam questions.

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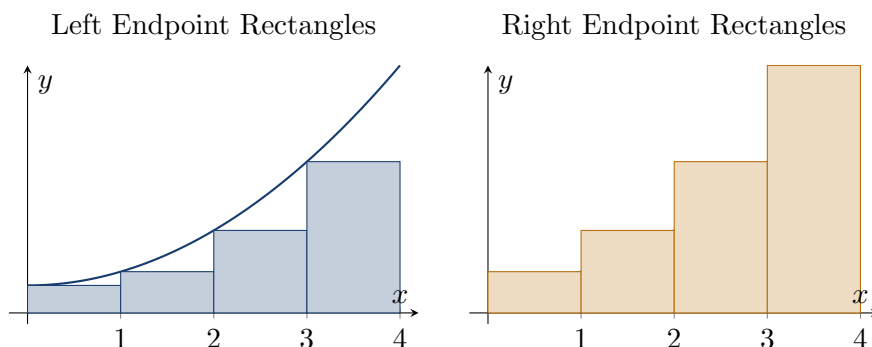
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1 Riemann Sums and Definite Integrals

1.1 The Basic Idea

Suppose you want to find the **area under a curve** $y = f(x)$ between $x = a$ and $x = b$. If f were a straight line, you could use simple geometry (triangles, trapezoids). But for a curved function, there is no direct formula – so we approximate.

The trick: chop the region into many thin vertical rectangles, find the area of each rectangle (easy: base $\times$ height), and add them all up. The more rectangles we use, the thinner they get, and the better our approximation becomes. As the number of rectangles $\rightarrow \infty$, the approximation becomes *exact* – this exact value is called the **definite integral**.



Notice: left endpoints underestimate the area for an increasing function, while right endpoints overestimate it. As rectangles get thinner, both approximations squeeze toward the same true value.

1.2 Partitions and Riemann Sums

Partition of an Interval

Divide $[a, b]$ into n subintervals of equal width

$$\Delta x = \frac{b - a}{n}, \quad x_i = a + i \Delta x, \quad i = 0, 1, \dots, n.$$

The points $x_0 = a, x_1, x_2, \dots, x_n = b$ form a **partition**. In each subinterval $[x_{i-1}, x_i]$, pick a sample point x_i^* (left endpoint, right endpoint, or midpoint).

Riemann Sum

$$R_n = \sum_{i=1}^n f(x_i^*) \Delta x$$

This is the sum of the areas (signed – negative if $f < 0$) of the n approximating rectangles.

- Left Riemann sum: $x_i^* = x_{i-1}$
- Right Riemann sum: $x_i^* = x_i$
- Midpoint Riemann sum: $x_i^* = \frac{1}{2}(x_{i-1} + x_i)$

1.3 The Definite Integral

Definite Integral

If f is continuous on $[a, b]$, the **definite integral** of f from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

This limit always exists (and gives the same value regardless of the sample points chosen) when f is continuous on $[a, b]$. Here a and b are the **limits of integration**, $f(x)$ is the **integrand**, and dx signals the variable of integration.

Key Note

The definite integral is a **number**, not a function. Geometrically, it represents the **net signed area** between the curve $y = f(x)$ and the x -axis on $[a, b]$: area above the axis counts positively, area below counts negatively.

Useful Summation Formulas

These help evaluate Riemann sums directly:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$$

1.4 Worked Examples

Example 1.1

Estimate the area under $f(x) = x^2$ on $[0, 2]$ using $n = 4$ rectangles and right endpoints.

Solution. $\Delta x = \frac{2-0}{4} = 0.5$. Right endpoints: $x_1 = 0.5, x_2 = 1, x_3 = 1.5, x_4 = 2$.

$$R_4 = 0.5[f(0.5) + f(1) + f(1.5) + f(2)] = 0.5[0.25 + 1 + 2.25 + 4] = 0.5(7.5) = 3.75$$

Example 1.2

Use the limit definition of the Riemann sum (right endpoints) to evaluate $\int_0^2 x^2 dx$.

Solution. $\Delta x = \frac{2}{n}, x_i = \frac{2i}{n}$.

$$\begin{aligned} R_n &= \sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \left(\frac{2i}{n} \right)^2 \frac{2}{n} = \frac{8}{n^3} \sum_{i=1}^n i^2 = \frac{8}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \\ &= \frac{8(n+1)(2n+1)}{6n^2} = \frac{4}{3} \cdot \frac{(n+1)(2n+1)}{n^2} \end{aligned}$$

Taking the limit as $n \rightarrow \infty$:

$$\int_0^2 x^2 dx = \lim_{n \rightarrow \infty} \frac{4}{3} \cdot \frac{2n^2 + 3n + 1}{n^2} = \frac{4}{3}(2) = \frac{8}{3}$$

Example 1.3

Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[3 + \frac{2i}{n} \right]^2 \frac{2}{n}$ as a definite integral.

Solution. Compare with the general Riemann sum on $[a, b]$: $\Delta x = \frac{2}{n}$ suggests $b - a = 2$, and $x_i = 3 + \frac{2i}{n}$ suggests $a = 3$. So $f(x) = x^2$ and the sum equals

$$\int_3^5 x^2 dx.$$

1.5 Practice Problems: Riemann Sums and Definite Integrals**Easy**

- (a) Using $n = 4$ and left endpoints, estimate $\int_0^4 (x + 1) dx$.
- (b) Write out (do not evaluate) the right Riemann sum R_5 for $f(x) = \sqrt{x}$ on $[1, 6]$.

Medium

- (a) Using midpoints with $n = 4$, approximate $\int_0^2 x^2 dx$ and compare with the exact value found above.
- (b) Evaluate $\int_1^3 (2x + 1) dx$ using the limit definition of the Riemann sum.

Hard

- (a) Using the limit definition, evaluate $\int_0^1 (x^3 - x) dx$.
- (b) Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \sqrt{1 + \frac{3i}{n}}$ as a definite integral (do not evaluate).

2 Properties of Integrals

2.1 The Basic Idea

Once we know what a definite integral represents (net signed area), several common-sense rules follow naturally. For instance, if you stretch a function vertically by a factor of 2, the area doubles. If you split a region at a point c between a and b , the total area equals the sum of the two pieces. These "obvious" facts are formalized as the **properties of integrals**, and they are the main tools used to simplify and evaluate integrals quickly without going back to the Riemann sum definition every time.

2.2 Core Properties

Basic Properties

Let f, g be integrable on the relevant interval, and c, k constants.

$$\int_a^a f(x) dx = 0 \quad (1)$$

$$\int_a^b k dx = k(b - a) \quad (2)$$

$$\int_b^a f(x) dx = - \int_a^b f(x) dx \quad (3)$$

$$\int_a^b kf(x) dx = k \int_a^b f(x) dx \quad (4)$$

$$\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx \quad (5)$$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx \quad (\text{for any } c) \quad (6)$$

Comparison Properties

- If $f(x) \geq 0$ on $[a, b]$, then $\int_a^b f(x) dx \geq 0$.
- If $f(x) \geq g(x)$ on $[a, b]$, then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$.
- If $m \leq f(x) \leq M$ on $[a, b]$, then $m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$.

Symmetry Shortcuts

For f integrable on $[-a, a]$:

- If f is **even** ($f(-x) = f(x)$): $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.
- If f is **odd** ($f(-x) = -f(x)$): $\int_{-a}^a f(x) dx = 0$.

These come from symmetry of area about the y -axis (even) or cancellation of positive/negative regions (odd).

2.3 Worked Examples

Example 2.1

Given $\int_0^5 f(x) dx = 8$ and $\int_0^3 f(x) dx = 3$, find $\int_3^5 f(x) dx$.

Solution. By the additivity property, $\int_0^5 f = \int_0^3 f + \int_3^5 f$, so

$$\int_3^5 f(x) dx = 8 - 3 = 5.$$

Example 2.2

Evaluate $\int_{-2}^2 (3x^5 - 4x^3 + 2x) dx$ using symmetry.

Solution. Every term is an odd power of x , so the integrand is an **odd function**. By the symmetry property,

$$\int_{-2}^2 (3x^5 - 4x^3 + 2x) dx = 0.$$

Example 2.3

Estimate bounds for $\int_1^4 \frac{1}{x} dx$ using the comparison property.

Solution. On $[1, 4]$, $\frac{1}{x}$ decreases from 1 to $\frac{1}{4}$, so $\frac{1}{4} \leq f(x) \leq 1$. Then

$$\frac{1}{4}(4-1) \leq \int_1^4 \frac{1}{x} dx \leq 1(4-1) \quad \Rightarrow \quad 0.75 \leq \int_1^4 \frac{1}{x} dx \leq 3.$$

(The exact value, $\ln 4 \approx 1.386$, indeed lies in this range.)

2.4 Practice Problems: Properties of Integrals

Easy

(a) If $\int_1^6 f(x) dx = 10$, find $\int_6^1 f(x) dx$.

(b) Evaluate $\int_2^7 5 dx$.

Medium

(a) Given $\int_0^4 f(x) dx = 6$ and $\int_0^4 g(x) dx = -2$, find $\int_0^4 [3f(x) - 2g(x)] dx$.

(b) Evaluate $\int_{-3}^3 (x^4 - 5x^2 + 1) dx$ given $\int_0^3 (x^4 - 5x^2 + 1) dx = 21.6$, using symmetry.

Hard

(a) Given $\int_0^2 f(x) dx = 5$, $\int_2^6 f(x) dx = -3$, and $\int_0^6 g(x) dx = 4$, evaluate $\int_0^6 [2f(x) - g(x)] dx$ (careful: relate the f piece across $[0, 6]$ first).

(b) Without evaluating, prove that $0 \leq \int_0^\pi \sin x dx \leq \pi$ using the comparison property.

3 The Fundamental Theorem of Calculus

3.1 The Basic Idea

This is the single most important result in this whole topic – it connects two ideas that seem completely different: **differentiation** (rate of change / slope) and **integration** (accumulated area). Before the Fundamental Theorem of Calculus (FTC), computing an integral meant taking a painful limit of a Riemann sum, as in Section 1. The FTC says: if you can find an **antiderivative** of f , you can evaluate the definite integral *instantly* by simple subtraction. It also tells us that “the derivative of an accumulated area function gives back the original function.” These are the two parts (Part 1 and Part 2) of the theorem.

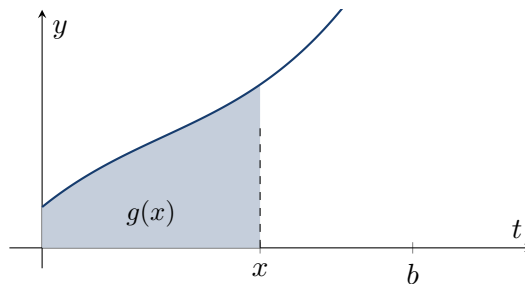
3.2 FTC Part 1: Differentiating an Integral

The Area Function

Let f be continuous on $[a, b]$. Define

$$g(x) = \int_a^x f(t) dt, \quad a \leq x \leq b.$$

$g(x)$ gives the accumulated (signed) area under f from a up to x . As x increases, this area accumulates – so g is itself a function of x .



FTC Part 1

If f is continuous on $[a, b]$ and $g(x) = \int_a^x f(t) dt$, then g is differentiable on (a, b) and

$$g'(x) = f(x).$$

In words: **the derivative of an integral (with variable upper limit) is the original integrand**, evaluated at that limit.

Chain-Rule Extension

If the upper limit is a function $u(x)$ instead of just x :

$$\frac{d}{dx} \int_a^{u(x)} f(t) dt = f(u(x)) \cdot u'(x).$$

If the lower limit is variable instead, flip the sign: $\frac{d}{dx} \int_{u(x)}^b f(t) dt = -f(u(x)) u'(x)$.

3.3 FTC Part 2: Evaluating Integrals

FTC Part 2 – The Evaluation Theorem

If f is continuous on $[a, b]$ and F is **any** antiderivative of f (i.e. $F' = f$), then

$$\int_a^b f(x) dx = F(b) - F(a) = \left[F(x) \right]_a^b.$$

Key Note

This is the formula you will use to evaluate almost every definite integral in this course: (1) find an antiderivative F , (2) plug in the upper limit, (3) subtract the value at the lower limit. No limits, no Riemann sums needed!

3.4 Worked Examples

Example 3.1

Find $g'(x)$ if $g(x) = \int_1^x \sqrt{t^2 + 1} dt$.

Solution. By FTC Part 1, $g'(x) = \sqrt{x^2 + 1}$ directly.

Example 3.2

Find $\frac{d}{dx} \int_2^{x^3} \sin(t^2) dt$.

Solution. Here $u(x) = x^3$, so by the chain-rule extension,

$$\frac{d}{dx} \int_2^{x^3} \sin(t^2) dt = \sin((x^3)^2) \cdot 3x^2 = 3x^2 \sin(x^6).$$

Example 3.3

Evaluate $\int_1^3 (2x^2 - 3) dx$ using FTC Part 2.

Solution. An antiderivative of $2x^2 - 3$ is $F(x) = \frac{2x^3}{3} - 3x$.

$$\int_1^3 (2x^2 - 3) dx = \left[\frac{2x^3}{3} - 3x \right]_1^3 = (18 - 9) - \left(\frac{2}{3} - 3 \right) = 9 - \left(-\frac{7}{3} \right) = \frac{34}{3}.$$

3.5 Practice Problems: Fundamental Theorem of Calculus

Easy

(a) Find $g'(x)$ if $g(x) = \int_0^x (t^3 + 1) dt$.

(b) Evaluate $\int_0^2 (3x^2 + 1) dx$.

Medium

(a) Find $\frac{d}{dx} \int_1^{x^2} \cos(t) dt$.

(b) Evaluate $\int_1^4 \left(\sqrt{x} + \frac{1}{x^2} \right) dx$.

Hard

(a) Find $\frac{d}{dx} \int_x^{x^2} e^{-t^2} dt$ (hint: split at a constant point, e.g. 0).

(b) If $F(x) = \int_2^x \frac{1}{1+t^2} dt$, find $F'(x)$ and determine whether F is increasing, decreasing, or constant on $\mathbb{R}$.

4 Antiderivatives and Indefinite Integrals

4.1 The Basic Idea

Differentiation takes a function and gives its rate of change. The reverse question is: given a rate of change, can we recover the **original function**? A function F is called an **antiderivative** of f if $F'(x) = f(x)$. For example, since $\frac{d}{dx}(x^2) = 2x$, we say x^2 is an antiderivative of $2x$. But so is $x^2 + 5$, or $x^2 - 100$ – because constants vanish under differentiation! This is why every antiderivative comes with a “ $+C$ ”.

4.2 Definitions

Antiderivative

F is an antiderivative of f on an interval I if $F'(x) = f(x)$ for all $x \in I$.

General Antiderivative Theorem

If F is one antiderivative of f on interval I , then **every** antiderivative of f on I has the form $F(x) + C$, where C is an arbitrary constant. This is because two functions with the same derivative on an interval differ only by a constant.

Indefinite Integral

The **indefinite integral** of f is the family of all antiderivatives, written

$$\int f(x) dx = F(x) + C.$$

Unlike the definite integral (a number), the indefinite integral is a **function** (family of functions).

4.3 Table of Basic Antiderivatives

Standard Antiderivative Formulas

Function	Antiderivative
k (constant)	$kx + C$
x^n ($n \neq -1$)	$\frac{x^{n+1}}{n+1} + C$
$\frac{1}{x}$	$\ln x + C$
e^x	$e^x + C$
a^x ($a > 0, a \neq 1$)	$\frac{a^x}{\ln a} + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\csc^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$
$\csc x \cot x$	$-\csc x + C$
$\frac{1}{\sqrt{1-x^2}}$	$\arcsin x + C$
$\frac{1}{1+x^2}$	$\arctan x + C$

Linearity of Antiderivatives

$$\int [kf(x) \pm g(x)] dx = k \int f(x) dx \pm \int g(x) dx$$

4.4 Worked Examples

Example 4.1

Find $\int (4x^3 - 6x + 5) dx$.

Solution. Apply the power rule term by term:

$$\int (4x^3 - 6x + 5) dx = 4 \cdot \frac{x^4}{4} - 6 \cdot \frac{x^2}{2} + 5x + C = x^4 - 3x^2 + 5x + C.$$

Example 4.2

Find $\int \left(3\sqrt{x} + \frac{2}{x} - \sec^2 x \right) dx$.

Solution. Rewrite $\sqrt{x} = x^{1/2}$:

$$\int 3x^{1/2} dx = 3 \cdot \frac{x^{3/2}}{3/2} = 2x^{3/2}, \quad \int \frac{2}{x} dx = 2 \ln|x|, \quad \int -\sec^2 x dx = -\tan x$$

$$\therefore \int \left(3\sqrt{x} + \frac{2}{x} - \sec^2 x \right) dx = 2x^{3/2} + 2 \ln|x| - \tan x + C.$$

Example 4.3

Find $f(x)$ given $f'(x) = 6x^2 - 4x$ and $f(1) = 5$ (an **initial value problem**).

Solution. General antiderivative:

$$f(x) = 2x^3 - 2x^2 + C.$$

Use the condition $f(1) = 5$: $2(1) - 2(1) + C = 5 \Rightarrow C = 5$. So $f(x) = 2x^3 - 2x^2 + 5$.

4.5 Practice Problems: Antiderivatives and Indefinite Integrals**Easy**

(a) Find $\int (5x^4 - 3x^2 + 2) dx$.

(b) Find $\int (\sin x + \cos x) dx$.

Medium

(a) Find $\int \left(\frac{3}{x} - 4e^x + \sqrt[3]{x} \right) dx$.

(b) Find $f(x)$ given $f'(x) = 3x^2 + 4x - 1$ and $f(0) = 2$.

Hard

(a) Find $\int \frac{x^3 - 2x + 1}{x} dx$ (hint: divide term by term first).

(b) Find $f(x)$ given $f''(x) = 12x^2 - 6x$, $f'(0) = 1$, $f(0) = 3$.

5 The Net Change Theorem

5.1 The Basic Idea

If you know your car's **velocity** at every instant, can you find how far you traveled (or your net displacement)? Yes – and this is exactly what the Net Change Theorem tells us. It is really just FTC Part 2 applied with a physical meaning: the integral of a **rate of change** equals the **total (net) change** in the original quantity over that time interval.

5.2 Statement and Interpretation

Net Change Theorem

If $F'(x) = f(x)$ is continuous on $[a, b]$, then

$$\int_a^b F'(x) dx = F(b) - F(a).$$

That is: the integral of a rate of change is the net change in the quantity, over $[a, b]$.

Common Applications

- If $v(t)$ is **velocity**: $\int_{t_1}^{t_2} v(t) dt = s(t_2) - s(t_1) = \text{net displacement}$.
- Total **distance travelled** (always positive) is $\int_{t_1}^{t_2} |v(t)| dt$ – split the interval where v changes sign.
- If $a(t)$ is **acceleration**: $\int_{t_1}^{t_2} a(t) dt = v(t_2) - v(t_1)$.
- If $R(t)$ is a **rate** (of water flow, population growth, cost, etc.): $\int_{t_1}^{t_2} R(t) dt = \text{total amount accumulated/changed}$.

5.3 Worked Examples

Example 5.1

A particle moves with velocity $v(t) = t^2 - 2t - 3$ (m/s) for $0 \leq t \leq 4$. Find the **net displacement** and the **total distance travelled**.

Solution. *Net displacement:*

$$\int_0^4 (t^2 - 2t - 3) dt = \left[\frac{t^3}{3} - t^2 - 3t \right]_0^4 = \left(\frac{64}{3} - 16 - 12 \right) - 0 = \frac{64}{3} - 28 = -\frac{20}{3} \text{ m.}$$

Total distance: First find where $v(t) = 0$: $t^2 - 2t - 3 = (t - 3)(t + 1) = 0 \Rightarrow t = 3$ (within $[0, 4]$).
 $v(t) < 0$ on $(0, 3)$ and $v(t) > 0$ on $(3, 4)$.

$$\int_0^3 (t^2 - 2t - 3) dt = \left[\frac{t^3}{3} - t^2 - 3t \right]_0^3 = 9 - 9 - 9 = -9$$

$$\int_3^4 (t^2 - 2t - 3) dt = \left(\frac{64}{3} - 16 - 12 \right) - (-9) = -\frac{20}{3} + 9 = \frac{7}{3}$$

Total distance = $|-9| + \left| \frac{7}{3} \right| = 9 + \frac{7}{3} = \frac{34}{3} \text{ m.}$

Example 5.2

Water flows into a tank at a rate of $R(t) = 5 + 2t$ litres/min. How much water enters the tank between $t = 0$ and $t = 10$ minutes?

Solution.

$$\int_0^{10} (5 + 2t) dt = \left[5t + t^2 \right]_0^{10} = (50 + 100) - 0 = 150 \text{ litres.}$$

5.4 Practice Problems: Net Change Theorem**Easy**

- (a) A particle has velocity $v(t) = 4t$ (m/s). Find its net displacement between $t = 0$ and $t = 3$.
- (b) A tank fills at rate $R(t) = 10$ L/min. Find the total volume added over 6 minutes.

Medium

- (a) An object has velocity $v(t) = 3t^2 - 12$ (m/s) on $[0, 4]$. Find the net displacement and the total distance travelled.
- (b) The rate of a chemical reaction is $R(t) = 6 - t$ grams/hour for $0 \leq t \leq 8$. Find the total amount of chemical produced.

Hard

- (a) A particle's acceleration is $a(t) = 6t - 4$ with $v(0) = 2$. Find the total distance travelled between $t = 0$ and $t = 3$.
- (b) Marginal cost of production is $C'(x) = 3x^2 - 24x + 48$ (taka per unit). Find the increase in cost when production rises from $x = 2$ to $x = 6$ units, and interpret whether cost is increasing throughout.

6 The Substitution Rule

6.1 The Basic Idea

Our antiderivative table (Section 4) only handles simple, direct functions. But what about $\int 2x \cos(x^2) dx$? This doesn't match any table entry directly – it is a **composite function** multiplied by (essentially) the derivative of its inner function. The **Substitution Rule** (u-substitution) is the integral version of the Chain Rule: we rename the "inner function" as a new variable u , rewrite the whole integral in terms of u , and (hopefully) it becomes a simple table integral.

6.2 The Rule

Substitution Rule – Indefinite Integrals

If $u = g(x)$ is differentiable and f is continuous on the range of g , then

$$\int f(g(x)) g'(x) dx = \int f(u) du.$$

Method – Step by Step

1. Choose $u = g(x)$ – usually the "inner function" (inside a root, power, trig, exponential, or denominator).
2. Compute $du = g'(x) dx$, and solve for dx (or for the exact expression appearing in the integral).
3. Rewrite the **entire** integral in terms of u only (no x should remain).
4. Integrate using the basic table.
5. Substitute back $u = g(x)$ to return to the original variable.

Substitution Rule – Definite Integrals

When the integral has limits, either:

- (a) Substitute back to x at the end and use the original limits, **or**
- (b) **Change the limits** directly: if $u = g(x)$, then when $x = a$, $u = g(a)$; when $x = b$, $u = g(b)$:

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

Method (b) is usually faster – no need to substitute back.

6.3 Worked Examples

Example 6.1

Evaluate $\int 2x \cos(x^2) dx$.

Solution. Let $u = x^2$, so $du = 2x dx$ – this matches exactly what's in the integral!

$$\int 2x \cos(x^2) dx = \int \cos u du = \sin u + C = \sin(x^2) + C.$$

Example 6.2

Evaluate $\int x\sqrt{4x^2 + 3} \, dx$.

Solution. Let $u = 4x^2 + 3$, so $du = 8x \, dx \Rightarrow x \, dx = \frac{du}{8}$.

$$\int x\sqrt{4x^2 + 3} \, dx = \int \sqrt{u} \cdot \frac{du}{8} = \frac{1}{8} \cdot \frac{u^{3/2}}{3/2} + C = \frac{1}{12}u^{3/2} + C = \frac{1}{12}(4x^2 + 3)^{3/2} + C.$$

Example 6.3

Evaluate $\int_0^1 3x^2 e^{x^3} \, dx$ (definite integral – change the limits).

Solution. Let $u = x^3$, so $du = 3x^2 \, dx$. When $x = 0$, $u = 0$; when $x = 1$, $u = 1$.

$$\int_0^1 3x^2 e^{x^3} \, dx = \int_0^1 e^u \, du = \left[e^u \right]_0^1 = e^1 - e^0 = e - 1.$$

Example 6.4

Evaluate $\int \tan x \, dx$.

Solution. Write $\tan x = \frac{\sin x}{\cos x}$. Let $u = \cos x$, so $du = -\sin x \, dx \Rightarrow \sin x \, dx = -du$.

$$\int \frac{\sin x}{\cos x} \, dx = \int \frac{-du}{u} = -\ln |u| + C = -\ln |\cos x| + C = \ln |\sec x| + C.$$

6.4 Practice Problems: Substitution Rule**Easy**

(a) Evaluate $\int (3x + 1)^5 \, dx$.

(b) Evaluate $\int \sin(4x) \, dx$.

Medium

(a) Evaluate $\int \frac{x}{\sqrt{1 - x^2}} \, dx$.

(b) Evaluate $\int_0^{\pi/2} \cos x \sin^3 x \, dx$ (change the limits).

Hard

(a) Evaluate $\int \frac{e^{2x}}{1 + e^{2x}} \, dx$.

(b) Evaluate $\int_1^2 \frac{\ln x}{x} \, dx$ (change the limits).

7 Answer Key

Brief final answers / solution outlines for all practice problems. Section numbers match the practice sets above.

7.1 Riemann Sums and Definite Integrals

1. Easy(a): $\Delta x = 1$; $L_4 = 1[f(0) + f(1) + f(2) + f(3)] = 1 + 2 + 3 + 4 = 10$.
2. Easy(b): $\Delta x = 1$; $R_5 = \sum_{i=1}^5 \sqrt{1+i} = \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{5} + \sqrt{6}$.
3. Medium(a): Midpoints 0.25, 0.75, 1.25, 1.75; $M_4 = 0.5(0.0625 + 0.5625 + 1.5625 + 3.0625) = 2.625$ (approx. of exact $8/3 \approx 2.667$).
4. Medium(b): Antiderivative style limit gives $\int_1^3 (2x+1)dx = [x^2 + x]_1^3 = 12 - 2 = 10$.
5. Hard(a): $\int_0^1 (x^3 - x)dx = \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_0^1 = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$.
6. Hard(b): $\int_1^4 \sqrt{x} dx$.

7.2 Properties of Integrals

1. Easy(a): -10 .
2. Easy(b): 25 .
3. Medium(a): $3(6) - 2(-2) = 22$.
4. Medium(b): Even function $\Rightarrow \int_{-3}^3 = 2(21.6) = 43.2$.
5. Hard(a): $\int_0^6 f = 5 + (-3) = 2$; answer $= 2(2) - 4 = 0$.
6. Hard(b): Since $0 \leq \sin x \leq 1$ on $[0, \pi]$, comparison gives $0(\pi) \leq \int_0^\pi \sin x dx \leq 1(\pi)$, i.e. $0 \leq \dots \leq \pi$.

7.3 Fundamental Theorem of Calculus

1. Easy(a): $g'(x) = x^3 + 1$.
2. Easy(b): $[x^3 + x]_0^2 = 10$.
3. Medium(a): $2x \cos(x^2)$.
4. Medium(b): $\left[\frac{2}{3}x^{3/2} - \frac{1}{x}\right]_1^4 = \left(\frac{16}{3} - \frac{1}{4}\right) - \left(\frac{2}{3} - 1\right) = \frac{16}{3} - \frac{1}{4} + \frac{1}{3} \approx 5.58$.
5. Hard(a): Split at 0: $\frac{d}{dx} \left[\int_x^0 + \int_0^{x^2} \right] = -e^{-x^2} \cdot (-1) \cdot (\text{sign fix}) \Rightarrow \text{result} = 2xe^{-x^4} + e^{-x^2}$.
6. Hard(b): $F'(x) = \frac{1}{1+x^2} > 0$ for all x , so F is strictly increasing on $\mathbb{R}$.

7.4 Antiderivatives and Indefinite Integrals

1. Easy(a): $x^5 - x^3 + 2x + C$.
2. Easy(b): $-\cos x + \sin x + C$.
3. Medium(a): $3 \ln|x| - 4e^x + \frac{3}{4}x^{4/3} + C$.
4. Medium(b): $f(x) = x^3 + 2x^2 - x + 2$.
5. Hard(a): $\int (x^2 - 2 + \frac{1}{x})dx = \frac{x^3}{3} - 2x + \ln|x| + C$.
6. Hard(b): $f'(x) = 4x^3 - 3x^2 + 1$; $f(x) = x^4 - x^3 + x + 3$.

7.5 Net Change Theorem

1. Easy(a): $\int_0^3 4t \, dt = 18 \text{ m}$.
2. Easy(b): $10 \times 6 = 60 \text{ L}$.
3. Medium(a): Net displacement $\int_0^4 (3t^2 - 12)dt = [t^3 - 12t]_0^4 = 64 - 48 = 16 \text{ m}$; zero of v at $t = 2$; distance $= |-16| + |16| = 32 \text{ m}$ (split at $t = 2$: $\int_0^2 = -16$, $\int_2^4 = 32$... total distance $= 16 + 32 = 48$ if computed piecewise – students should verify by splitting explicitly).
4. Medium(b): $\int_0^8 (6 - t)dt = [6t - \frac{t^2}{2}]_0^8 = 48 - 32 = 16 \text{ g}$.
5. Hard(a): $v(t) = 3t^2 - 4t + 2$ (always positive, check discriminant < 0), so distance = displacement $= \int_0^3 v \, dt = [t^3 - 2t^2 + 2t]_0^3 = 27 - 18 + 6 = 15 \text{ m}$.
6. Hard(b): $\int_2^6 (3x^2 - 24x + 48)dx = [x^3 - 12x^2 + 48x]_2^6 = (216 - 432 + 288) - (8 - 48 + 96) = 72 - 56 = 16$ taka; check sign of $C'(x)$ on $[2, 6]$ to discuss monotonicity (it dips to 0 at $x = 4$).

7.6 Substitution Rule

1. Easy(a): $\frac{1}{18}(3x + 1)^6 + C$.
2. Easy(b): $-\frac{1}{4}\cos(4x) + C$.
3. Medium(a): $-\sqrt{1 - x^2} + C$.
4. Medium(b): $u = \sin x$: $[\frac{u^4}{4}]_0^1 = \frac{1}{4}$.
5. Hard(a): $u = 1 + e^{2x}$: $\frac{1}{2}\ln(1 + e^{2x}) + C$.
6. Hard(b): $u = \ln x$: $[\frac{u^2}{2}]_0^{\ln 2} = \frac{(\ln 2)^2}{2}$.

8 Important Questions for the 1-1 Semester Exam

High-Yield Topics

Based on typical CSE 1-1 calculus exam patterns in Bangladeshi universities, prioritize these:

1. Evaluate a definite integral using the **limit definition of the Riemann sum** (very common short/long question) – practice $\int x^2 dx$ and $\int x^3 dx$ style problems using $\sum i$, $\sum i^2$, $\sum i^3$ formulas.
2. State and prove/apply **FTC Part 1 and Part 2**; differentiate integrals with variable limits (including composite upper limits like x^2 or $\sqrt{x}$).
3. Evaluate definite and indefinite integrals using **substitution**, including changing limits directly.
4. Apply **properties of integrals** (linearity, additivity, odd/even symmetry) to compute integrals from given data, without directly evaluating.
5. **Net Change Theorem** word problems: displacement vs. total distance travelled (a favorite trap question – always check where $v(t) = 0$).
6. Solve **initial value problems**: given $f'(x)$ (or $f''(x)$) and initial condition(s), find $f(x)$.
7. Prove the **comparison property** / establish bounds for an integral without evaluating it exactly.
8. Mixed problem: compute area under a curve using both the Riemann sum definition *and* FTC, and compare/verify the answers match.
9. Antiderivative table recall: be fluent with all trig, exponential, and power-rule antiderivatives – these appear inside larger substitution problems.
10. Conceptual/short questions: difference between definite and indefinite integral; why “+C” is needed; geometric meaning of $\int_a^b f(x) dx$ as signed area.

Exam Strategy

- Always double-check whether a question wants **net displacement** or **total distance** – read carefully.
- For substitution problems, always confirm no leftover x terms remain before integrating in u .
- Show the antiderivative $F(x)$ explicitly before plugging in limits – partial marks are given for correct antiderivatives even with an arithmetic slip.
- Practice mental arithmetic for evaluating $F(b) - F(a)$ quickly and accurately under exam time pressure.